CS-104 Retrieval Quiz

Fall 2026 · Week 6, Friday

Name: _____

Closed book, no devices. For questions 1–3, circle the one best answer. For question 4, write your code in the space provided (no algorithm, documentation, or test cases needed).

1.

What does this code print?

```
def half(x):
    return x / 2

# --- main program ---
total = half(10) + half(4)
print(total)
```

a) `7`
b) `7.0`
c) `14.0`
d) `None`

2.

What is wrong with this function?

```
def fahrenheit(celsius)
    return celsius * 9 / 5 + 32
```

a) It should use `print` instead of `return`
b) `celsius` must be assigned a value before the function is defined
c) Nothing; it is correct
d) The `def` line is missing a colon (`:`) at the end

3.

Assume a function `cube(x)` has been defined that returns `x` cubed. What does `print(cube(1 + 2 + 3))` print?

a) `36`
b) `6`
c) `216`
d) `18`

4.

A cylinder's volume is pi × radius² × height. Write a function `cylinder_volume(radius, height)` that returns the volume, using `math.pi` (assume `import math` was already done). Then call it for a radius of 2 and a height of 10, store the result in a variable, and print it rounded to 1 decimal place.